Step 7 : Make decision on H_0 choosing the suitable rejection rule from the following table corresponding to H_1.

Alternative Hypothesis (H_1)	$\mu_X \neq \mu_Y$	$\mu_X > \mu_Y$	$\mu_X < \mu_Y$
Rejection Rule	$\lvert z_0 \rvert \geq z_{\alpha/2}$	$z_0 > z_\alpha$	$z_0 < -z_\alpha$

Example 1.11

A Model Examination was conducted to XII Standard students in the subject of Statistics. A District Educational Officer wanted to analyze the Gender-wise performance of the students using the marks secured by randomly selected boys and girls. Sample measures were calculated and the details are presented below:

Gender	Sample Size	Sample Mean	Sample Standard deviation
Boys	100	50	4
Girls	150	51	5

Test, at 5% level of significance, whether performance of the students differ significantly with respect to their gender.

Solution:

Step 1 : Let μ_X and μ_Y denote respectively the average marks secured by boys and girls in the Model Examination conducted to the XII Standard students in the subject of Statistics. Then, the null and the alternative hypotheses are

Null hypothesis: $H_0: \mu_X = \mu_Y$

i.e., there is no significant difference in the performance of the students with respect to their gender.

Alternative hypothesis: $H_1 : \mu_X \neq \mu_Y$

i.e., performance of the students differ significantly with the respect to the gender. It is a two-sided alternative hypothesis.

Step 2 : **Data**

The given sample information are

Gender of the Students	Sample Size	Sample Mean	Sample Standard Deviation
Boys	$m = 100$	$\overline{x} = 50$	$s_X = 4$
Girls	$n = 150$	$\overline{y} = 51$	$s_Y = 5$

Since $m \geq 30$ and $n \geq 30$, both the samples are large.

Step 3 : **Level of significance**

$\alpha = 5\%$

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